

# IPsec VPN Technical & Forensic Audit Report

Target File: demo-aes128-ikev2-tunnel.pcap | Session: sess\_20260901\_6b66b5 | Timestamp: 2026-09-01T13:40:49

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## 1. Capture & Tunnel Forensics

- Packets Analyzed: 62 (IKE: 2, ESP: 60, Other: 0)
- Captured Duration: 0.61s | Size: 9.5 KB
- Protocol: IPsec (IKEv2) | Mode: Tunnel Mode

## 2. Cryptographic Security Assessment

- Encryption Cipher: AES-128-CBC
- Integrity / Authentication: HMAC-SHA256
- Diffie-Hellman Group: MODP-2048 (Group 14)
- Perfect Forward Secrecy: Enabled | Replay Protection: Enabled

## 3. AI Traffic Classification

- Predicted Application: Web Browsing (65% Confidence)
- Overall Security Score: 91/100 (EXCELLENT) | Risk: 14/100 (LOW)

## 4. Detected Threat Findings

- [MEDIUM] Moderate Diffie-Hellman Group Strength: Upgrade DH Group to MODP-3072 (Group 15) or ECP-384 (Group 20)
- [LOW] CBC Cipher Mode Utilization: Transition to AEAD mode such as AES-128-GCM or AES-256-GCM
- [INFO] Unencrypted Endpoint IP Metadata: Consider routing traffic through an anonymizing gateway or overlay if required